Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building distributed services, full-stack apps, and CI/CD using Java, Python, React, Node.js, AWS, and Kubernetes. Mentored engineers in Agile teams.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: React, Node.js, Express.js, Django, Flask, FastAPI, HTML/CSS, REST APIs, Microservices, Distributed Systems.

Databases: PostgreSQL, MySQL, DynamoDB, Redis.

Developer Tools: Git, Jenkins, GitLab CI/CD, AWS, Azure, GCP, Docker, Kubernetes, Helm, Terraform, Linux, Bash.

Testing & Practices: Agile/Scrum, JUnit, Pytest, Unit Testing, Object-Oriented Design, Design Patterns, Data Structures.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault, improving scalability and manageability of the deployment architecture.
- Built self-service developer platform (dashboard and CLI/YAML automation) adopted by **15 engineers**, cutting environment start, stop, and delete time by up to **90%**.
- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** while enabling reliable rollback and recovery.
- Served as Quality Coach and SonarQube Specialist, enforcing coding standards for maintainability and raising automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, FastAPI, React, PostgreSQL, Docker] Jul 2026

- Built end-to-end ML forecasting system for NYC taxi demand across **263 zones** (**2.3M-row** dataset), serving predictions via FastAPI REST API to a React dashboard with PostgreSQL and Redis.
- Designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over Random Forest.

PolicyPilot | Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Built full-stack TypeScript healthcare app (**React SPA, Express 5 REST API**, MCP server) unifying insurance prior-authorization workflows, with Zod schema validation at every data boundary.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).